

3623. Count Number of Trapezoids I

Difficulty: Medium

Link: <https://leetcode.com/problems/count-number-of-trapezoids-i/description/?envType=daily-question&envId=2025-12-02>

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Problem:

You are given a 2D integer array `points`, where `points[i] = [xi, yi]` represents the coordinates of the i^{th} point on the Cartesian plane.

A **horizontal trapezoid** is a convex quadrilateral with **at least one pair** of horizontal sides (i.e. parallel to the x-axis). Two lines are parallel if and only if they have the same slope.

Return the *number of unique **horizontal trapezoids*** that can be formed by choosing any four distinct points from `points`.

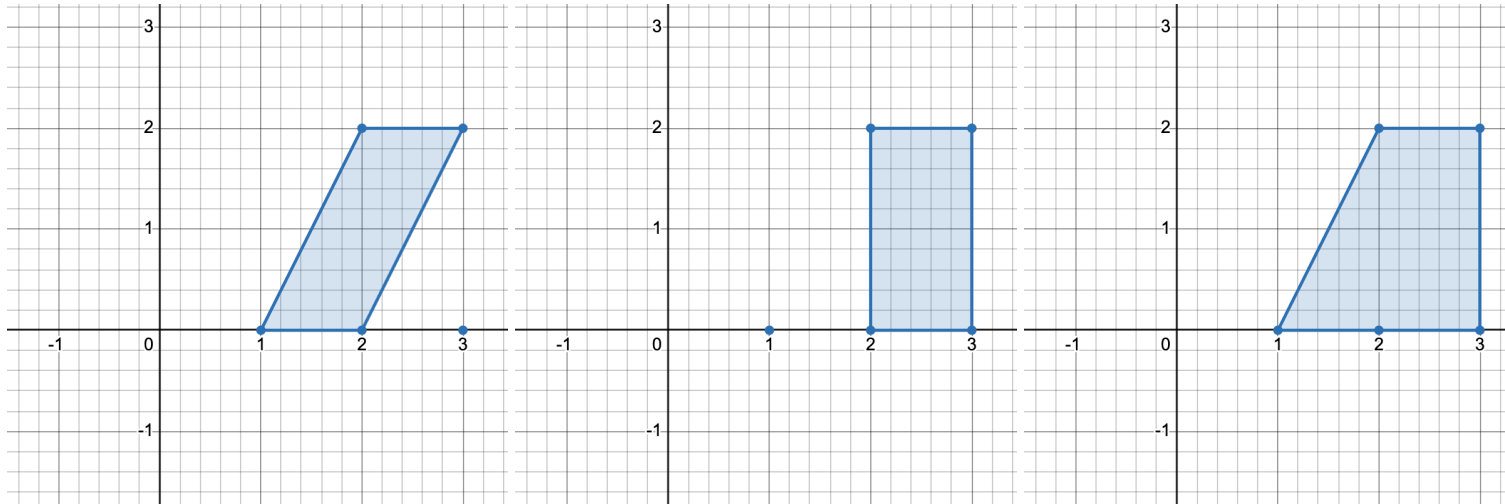
Since the answer may be very large, return it **modulo** $10^9 + 7$.

Example 1:

Input: `points = [[1,0],[2,0],[3,0],[2,2],[3,2]]`

Output: 3

Explanation:



There are three distinct ways to pick four points that form a horizontal trapezoid:

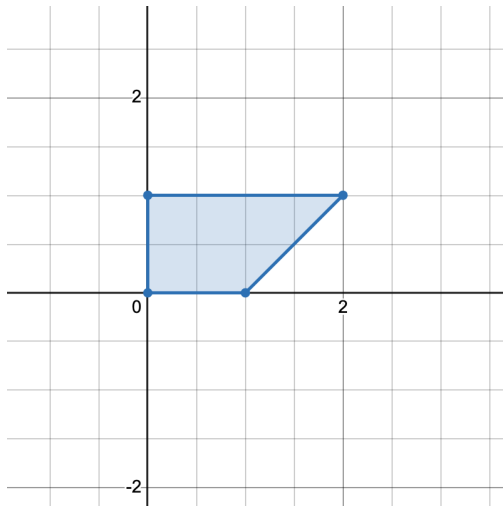
- Using points $[1,0]$, $[2,0]$, $[3,2]$, and $[2,2]$.
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Example 2:

Input: points = $[[0,0],[1,0],[0,1],[2,1]]$

Output: 1

Explanation:



There is only one horizontal trapezoid that can be formed.

Constraints:

- $4 \leq \text{points.length} \leq 10^5$
- $-10^8 \leq x_i, y_i \leq 10^8$
- All points are pairwise distinct.